

# Question Paper

Exam Date & Time: 05-Jan-2026 (02:00 PM - 05:00 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026

SUBJECT: CSE 3122/CSE\_3122 SOFTWARE ENGINEERING

(SPL: COMPUTER SCIENCE & ENGINEERING)

SOFTWARE ENGINEERING [CSE 3122]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Identify a real-world situation where an organization can benefit from using a software product line and briefly explain how it would be applied. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Determine which Software Development Life Cycle (SDLC) model is used in Automotive Software Development and explain the Model with a diagram. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Identify which Life cycle model is used in developing the Software for Safety Critical devices like MRI Scanners. Explain why this model is considered to be a meta model. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) Imagine a university plan to automate its course registration system, which currently depends on paper forms manually entered by staff. Students need to register online, select courses, and receive confirmation emails, while administrators must verify eligibility and generate timetables. As a system analyst, explain how you would apply structured analysis to model this system using Data Flow Diagrams (DFDs) up to Level 1, clearly identifying key processes, data stores, and data flows. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Assume a software company is developing an online examination system for a university. The draft SRS includes functional requirements such as user login and test scheduling but fails to specify constraints on response time, concurrent users, and security measures. As requirements engineer, explain how you would revise the SRS to include clear non-functional requirements, ensuring the system meets both user expectations and performance standards. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Suppose a student information system has a single module responsible for attendance, fee calculation, and grade entry. Identify the type of cohesion demonstrated here and suggest how you would improve the module's design. (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3A) An Automatic Resume Screening System is designed to help companies automate their hiring process. Recruiters can upload job descriptions and review shortlisted candidates. Job seekers can

create profiles, upload resumes, and track their application status. The system uses AI to compare candidate skills with job requirements and rank them. The admin manages user accounts, recruitment analytics, and system updates.

Based on the above scenario, draw a Use Case Diagram showing the actors (Job Seeker, Recruiter, Admin, AI Engine), use cases, and relationships (include / extend / generalization where needed).

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3B) A Smart Home Energy Management System monitors electricity usage of appliances and allows homeowners to control devices remotely via a mobile app. The system also provides weekly energy consumption reports and notifies users when usage exceeds limits. Draw a Level 0 DFD (Context Diagram) and Level 1 DFD for this system, showing external entities, main processes, and data flows and decomposition of the main process in Level 1. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) Distinguish between Model-Oriented Methods and Property-Oriented Methods in formal specification, giving one example for each. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4A) A Central Library is building an automated book checkout process. Students initiate the process by scanning their Id cards at self-service Kiosk machine. Kiosk machine sends student Id number to the Library Management System (LMS) to check if card is invalid or maximum books borrowed. If there is any such error, LMS responds with appropriate error message. If there is no error, student can further scan the book barcode. Kiosk machine sends the barcode and student Id number to LMS. If the book is not reserved, LMS will reserve the book, update due date and respond with appropriate confirmation message. If the book is reserved by another student, LMS responds with appropriate error message. Draw a UML Sequence Diagram for the above process. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) Design an activity diagram for computing a Bill in a Restaurant Management System. There should be a charge for each item delivered. The total amount should be subject to tax and a service charge of 18% for groups of six or more. For smaller groups, there should be a blank entry for a gratuity (Tip) according to the customer's discretion. Any coupons or gift certificates submitted by the customer should be subtracted. (3)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 4C) Stock Market Monitoring System notifies multiple independent displays whenever a stock price changes by Stock data object, which is not aware of specific details of Displays. Identify a Design Pattern to implement one-to-many dependency between stock data and displays. Explain how this pattern achieves decoupling between them. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5A) Write a program in any programming language to check whether a given number is prime or not and then perform software test coverage analysis on the code by determining the Statement Coverage, Branch Coverage, and Condition Coverage. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) A software company frequently updates its online banking application with new features like fund scheduling and bill payments. Explain the purpose of regression testing in this scenario. Why is it important to perform regression testing after every new feature addition? (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) A software development team is integrating three key modules, User Authentication (A), Order Processing (B), and Inventory Management (C), into a new e-commerce system. Module A is complete and tested. Modules B and C have minor dependencies on A, but A does not depend on B or C. Module B has critical dependencies on C to check stock before finalizing an order. The team has decided to adopt an Incremental Integration Testing strategy. Given the module dependencies and development status, Analyze and justify whether to use a Top-Down or Bottom-Up incremental integration approach for this specific system. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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| Event Name | Subject |
|------------|---------|
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE\_3122:Software Engineering

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 10        |
| 2            | 2              | CO 2        | 5         |
| 3            | 3              | CO 3        | 10        |
| 4            | 4              | CO 4        | 15        |
| 5            | 5              | CO 5        | 10        |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
|------------|---------|

058-MIT BLR-MAKEUP EXAM-2025-2026 CSE\_3122:Software Engineering

| #     | Level       | No of Questions | Marks (%) |
|-------|-------------|-----------------|-----------|
| 1     | Remember    | 0               | 0(0%)     |
| 2     | Understand  | 0               | 0(0%)     |
| 3     | Application | 7               | 24(48%)   |
| 4     | Analysis    | 7               | 23(46%)   |
| 5     | Create      | 1               | 3(6%)     |
| 6     | Evaluate    | 0               | 0(0%)     |
| Total |             | 15              | 50 (100%) |